ChiUnited Church · Tech Ministry · Saturday, October 24, 2026, 2 to 5 PM

Church Tech 101

Session one: why church tech exists, a right strategy for it, and how to read a signal chain.

Dr. Bill Yomes

I. What is the purpose of church tech?

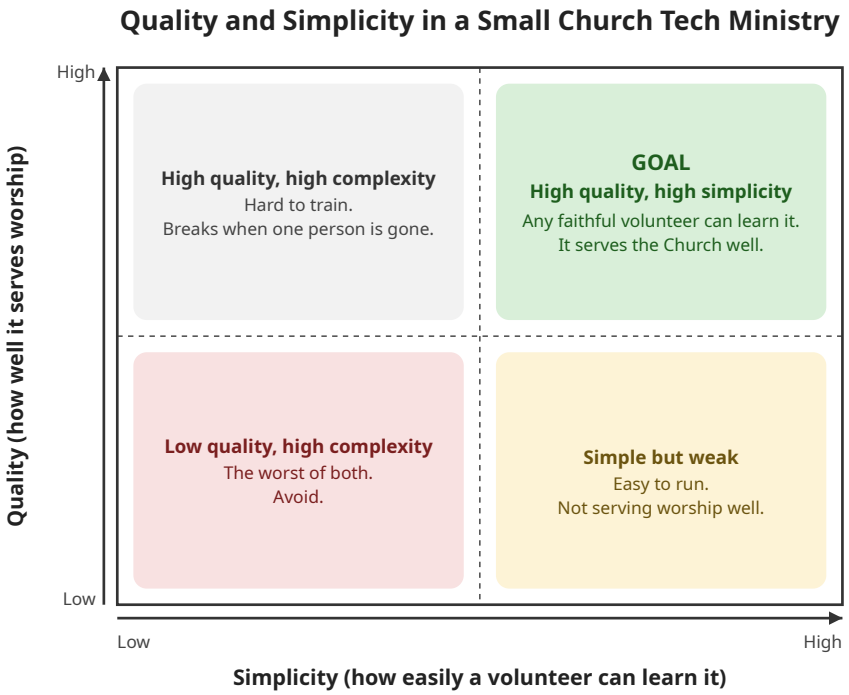
A. Glorify God
B. Enable disciple-making
C. Facilitate worship services and events
D. Grow the tech ministry

II. What is a right strategy for church tech?

A. **Quality through simplicity.**

> Church tech systems should be simple enough for any faithful volunteer to learn and strong enough to serve the Church well.

B. **The quality versus complexity matrix.** The goal is the top right square: high quality and high simplicity.



C. **Growth through simplicity.** A system that is easy to learn is easy to teach, so more people can serve and the ministry grows.

III. The signal chain

Most church tech can be understood through the signal chain. Every chain has three parts:

1. **Source:** where the signal starts.

2. **Chain:** everything it passes through.
3. **Endpoint:** where it ends and someone sees or hears it.

Example: a solo guitar player

Source: guitar → **Cable** → **Amplifier** → **Endpoint: speaker in the amp**

Even this small setup has a source, a chain and an endpoint. A church system is the same idea with more links.

Signal chains at ChiUnited

1. The audio chain (basic)

Source → **Cables** → **Sound board** → **Amplifier** → **Speakers**

The sound board also sends a second feed, to the livestream:

Sound board → **Video input box** → **Computer** → **Livestream**

2. The video chain (basic)

Cameras → **Video input box** → **Computer** → **Livestream**

3. The lyrics and screen chain (basic)

The lyrics computer sends to three places:

Computer → **Two congregation screens**

Computer → **Confidence monitor**

Computer → **Video input box** → **Computer** → **Livestream**

At ChiUnited the "sound board" is the Behringer XR18, the "video input box" is the Blackmagic ATEM Mini Pro, and the livestream computer is the Mac mini running OBS.

IV. The exercise: signal chain discovery

Each team member goes to their usual spot. Then we call out numbers from source to endpoint, tracing the signal chains through the sanctuary. This gives every team member a better understanding of where they are in the chain and how the tech as a whole serves the church.

1. Everyone goes to their usual spot (stage, rack, sound board, video, lyrics computer, streaming computer).
2. The leader calls "One" for the source of a chain. The person there says what they are, and what comes in and goes out.
3. Call the numbers in order, link by link, until the endpoint.
4. Repeat for the audio, video and lyrics chains. Note where a chain splits.

Worksheet. Fill it in as the numbers are called.

Chain	#	Station	Who is here	What comes in	What goes out
Audio	1	Source (mic, instrument)			
	2	Cables, stage box, snake			
	3	Sound board			
	4	Amplifier			
	5	Speakers			
Audio to stream	3b	Sound board to video input box			
	4b	Video input box to computer			
	5b	Livestream			
Video	1	Cameras			
	2	Video input box			
	3	Computer			
	4	Livestream			
Lyrics	1	Lyrics computer			
	2a	Congregation screens			
	2b	Confidence monitor			
	2c	Video input box			
	3c	Computer			
	4c	Livestream			

Prepared for the ChiUnited tech team by Dr. Bill Yomes. Version of Friday, October 23, 2026.

Live version: wtsn.me/chi/train